Shuai Zhang

Senior Staff Engineer & Manager, Computer Vision — Qualcomm Multimedia R&D
On-Device Agentic AI · Multimodal Perception · VLM/VLA · Edge AI
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Summary

Senior computer-vision researcher with 8+ years of experience designing and shipping perception algorithms on edge devices, currently leading R&D of on-device agentic AI pipelines for smart glasses, AI PCs, and IoT systems. Deep expertise in real-time segmentation, detection, multimodal perception, VLM/VLA inference, and on-device model optimization, with a track record of deploying flagship vision features on Snapdragon chipsets. Ph.D. in Mathematics (UC Irvine); foundation in numerical optimization and deep-learning theory.

Experience

Senior Staff Engineer & Manager, Computer Vision — Qualcomm Technologies 11/2024 – Present

- Lead R&D of on-device agentic AI pipelines — multimodal perception, VLM/LLM reasoning, planning, and tool use — deployed on smart glasses, AI PCs, and IoT systems under tight latency and power budgets.
- Designed end-to-end on-device stack: open-vocabulary tagging, detection, segmentation, VLM/LLM reasoning, multimodal embedding-based retrieval and memory, and efficient model inference.
- **Agentic AI (ForeSea)**: designed an agentic multimodal RAG pipeline for long-horizon video understanding, combining person-centric tracking, multimodal retrieval, and VideoLLM reasoning. Integrated as product on Qualcomm IoT Insight Platform (QIP).
- **Smart Glasses**: architected an end-to-end personalized on-device AI assistant for smart glasses, integrating ASR, detection, segmentation, RAG, and agentic reasoning across glasses + companion compute.
- **VLA (Latent Bridge)**: research on dual-system VLA inference acceleration. Proposed feature-delta prediction in latent space, alternately invoking a shallow vision encoder on redundant frames; reduces VLM calls by 50–75% while retaining 95–100% task success, delivering 1.65–1.73× end-to-end speedup on two architecturally distinct VLAs: GR00T-N1.6 (feature-space bridge) and $Pi_{0.5}$ (KV-cache bridge).

Staff Engineer (Machine Learning) — Qualcomm Technologies 11/2020 – 11/2024

- **Limitless Segmentation** on Snapdragon 8 Elite: designed an in-house, real-time multimodal-prompt segmentation model (text / audio / point / box prompts) deployed as flagship semantic-segmentation feature in the Qualcomm camera pipeline.
- **Segmentation-based Cognitive ISP** (AI camera): architected the segmentation-driven ISP pipeline with a shared encoder serving semantic segmentation, person/pet detection, instance depth, and scene classification. Recognized with the **2023 Edge AI & Vision Product of the Year Award** (525K+ Google results).
- Led full on-device model optimization pipeline: quantization, pruning, and low-latency inference on Qualcomm AI Hub and mobile SoCs.

Senior Engineer (Machine Learning) — Qualcomm Technologies 07/2017 – 11/2020

- Designed production **biometric perception** algorithms (face detection, anti-spoofing, gaze/attention estimation, on-device user adaptation) and camera segmentation features.
- Build in-house model training pipeline on top of Caffe and then Pytorch, including quantization, pruning, distillation, multi-tasks training and evaluation.

Data Scientist Intern — Zillow Group 06/2016 – 09/2016

- Built recommendation algorithms and contributed to Zestimate (home-price estimation) modeling pipeline.

Selected Publications (full list: https://zsivine.github.io/research)

- **DiSa: Saliency-Aware Foreground-Background Disentangled Framework for Open-Vocabulary Semantic Segmentation.** arXiv:2601.20064, 2026.
- **ForeSea: AI Forensic Search with Multi-modal Queries for Video Surveillance.** ECCV 2026.
- **Latent Bridge: Feature Delta Prediction for Efficient Dual-System Vision-Language-Action Model Inference.** arXiv:2605.02739, 2026.

- **SEMA: A Scalable and Efficient Mamba-like Attention via Token Localization and Averaging.** ICML 2026.
- **MobileInst: Video Instance Segmentation on the Mobile.** AAAI 2024.
- **AutoShuffleNet: Learning Permutation Matrices via an Exact Lipschitz Continuous Penalty in Deep CNNs.** KDD 2020 — Most Influential KDD Paper.
- **TRP: Trained Rank Pruning for Efficient Deep Neural Networks.** IJCAI 2020.
- **Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets.** ICLR 2019.
- **DAC: Data-free Automatic Acceleration of Convolutional Networks.** IEEE WACV 2019 — co-first author.
- **BinaryRelax: A Relaxation Approach for Training DNNs with Quantized Weights.** SIAM Journal on Imaging Sciences, 2018 — co-first author.

Selected Patents

- Semantic image segmentation via depth-fused cross-attention transformer. *US 12,141,981* — granted Nov 2024.
- Scene segmentation and object tracking. *US 12,236,614* — granted Feb 2025.
- User adaptation for biometric authentication. *US 11,887,404* — granted Jan 2024.
- Personalized eye-openness estimation. *US App. 16/239,352* — pending.

Awards & Recognition

- **Invited Panelist**, CVPR Workshop on Computer Vision with Small Data (cv4small) 2026
- **Primary Organizer**, CVPR Workshop on Efficient Deep Learning for Computer Vision (ECV) 2025 & 2026
- **Primary Organizer**, IEEE Low Power Computer Vision Challenge — `www.lpcv.ai` 2025 & 2026
- **IEEE Senior Member** 2023
- **Edge AI & Vision Product of the Year** — Segmentation-based Cognitive ISP (Qualcomm) 2023
- **3rd Place**, Low-Latency Object Detection — IEEE LPIRC 2019
- **2nd Place**, Image Classification & Detection — IEEE LPIRC 2018
- **Kovalevsky Outstanding Ph.D. Thesis Award** — UC Irvine 2017
- **SIAM Student Travel Award** 2016
- **Von Neumann Outstanding Research Award** — UC Irvine 2015

Education

University of California, Irvine Irvine, CA

- *Ph.D., Mathematics* 2017
  Advisor: Prof. Jack Xin | Research: Compressed Sensing, Image Processing, Machine Learning
- *M.S., Mathematics* 2014

Shandong University Jinan, China

- *M.S., Computational Mathematics* 2012
- *B.S., Applied Mathematics* 2009